

MAT61-4

Stochastic Calculus

Albert School — 30 hours

Preface

This course is where three earlier threads of your mathematical education are finally braided together. In **Discrete Stochastic Models** (MAT21-4) you studied random walks, Markov chains with their transition matrices and stationary distributions, and the martingale — the “fair game” whose expected future value, given the present, is exactly the present. There the time was discrete: the walker stepped at $t = 0, 1, 2, \dots$. In **Probability & Statistics II** (MAT31-2) you crossed from discrete to **continuous** randomness — densities, the normal distribution, the Central Limit Theorem, and Monte Carlo simulation of expectations. And in **Differential Equations** (MAT41-4) you learned to read a law of motion $y' = f(t, y)$ as a recipe that, from an initial condition, pins down an entire trajectory, and to solve such equations both by hand and numerically with Euler’s method.

Stochastic calculus is what you get when you let the differential equation be **driven by noise**. Imagine the deterministic dynamics of MAT41-4 with a random term added at every instant: a stock price that drifts upward on average but is buffeted continuously by news, a particle of pollen that jitters under molecular bombardment. The mathematical engine of that continuous noise is **Brownian motion**, the continuous-time limit of the random walk you already know. The catch — and the reason this subject needs its own calculus — is that a Brownian path is so violently jagged that it has **no derivative anywhere**. The symbol dW cannot mean “the rate of change of W ”, because there is no rate of change. Classical calculus breaks down at the first step.

The resolution is a single, beautiful idea that you should keep in view from the opening page: for a Brownian path the increment dW over a time dt is so large — of size $\sqrt{dt}$ rather than dt — that its **square** cannot be neglected. Where Newton and Leibniz threw away $(dx)^2$ as negligibly small, we must keep $(dW)^2$, and it obeys the astonishing rule

$$(dW)^2 = dt.$$

Every formula in this book that differs from ordinary calculus — Itô’s correction term, the Black–Scholes equation, the $\sqrt{\Delta t}$ scaling of a simulation — is a direct consequence of that one line.

We travel a single arc. We begin by recalling **stochastic processes** and sharpening the martingale and filtration language. We build **Brownian motion** and establish its quadratic variation. We then construct the **Itô integral** $\int H dW$ — the only honest way to integrate against a non-differentiable path — and prove its two defining properties, the zero-expectation and the Itô isometry. From there **Itô’s formula** gives us a working chain rule, and with it we solve **stochastic differential equations** — geometric Brownian motion, the mean-reverting Ornstein–Uhlenbeck process, the CIR model. The pay-off is **Black–Scholes**: we derive the option-pricing equation by combining Itô’s formula with a hedging argument, value options under the risk-neutral measure, and read off the Greeks. We close where computation takes over from formula — **Monte Carlo pricing** by Euler–Maruyama simulation, with the variance-reduction tricks that make it practical.

Keep the refrain in mind throughout: **square the increment**. It is the whole course told in three words.

Contents

Preface	2
1 Stochastic processes	4
1.1 A concrete process first: the random walk	4

1.2	Filtrations: information revealed over time	4
1.3	Markov chains, recalled and put in continuous-time context	5
1.4	Martingales: the formal fair game	5
2	Brownian motion	6
2.1	From random walk to Brownian motion	6
2.2	The definition	7
2.3	The covariance function	7
2.4	Quadratic variation: the heart of the matter	8
2.5	Why Brownian paths are nowhere differentiable	9
3	Itô integration	9
3.1	Building the integral for simple integrands	9
3.2	The zero-expectation property	10
3.3	The Itô isometry	11
4	Itô's formula	12
4.1	The need for a corrected chain rule	12
4.2	Itô's formula for a function of Brownian motion	12
4.3	The general Itô process and its formula	13
4.4	Computing expectations with Itô	14
5	Stochastic differential equations	14
5.1	Drift, diffusion, and what a solution means	14
5.2	Existence and uniqueness	15
5.3	Geometric Brownian motion	15
5.4	The Ornstein–Uhlenbeck process and mean reversion	16
5.5	The CIR model	17
6	The Black–Scholes model	18
6.1	The setup: stock, bond, and option	18
6.2	Deriving the Black–Scholes PDE by delta hedging	18
6.3	Risk-neutral valuation	19
6.4	The Black–Scholes formula	20
6.5	The Greeks	20
6.6	Put–call parity	21
7	Monte Carlo option pricing	21
7.1	Euler–Maruyama discretization	21
7.2	The Monte Carlo estimator and its confidence interval	22
7.3	Variance reduction	23
8	Epilogue: one rule, told seven ways	23

1 Stochastic processes

A **stochastic process** is simply a collection of random variables indexed by time, $(X_t)_{t \geq 0}$ (continuous time) or $(X_n)_{n \in \mathbb{N}}$ (discrete time). You have met several already; this chapter gathers them, fixes vocabulary, and introduces the two ideas — **filtrations** and **martingales in their formal sense** — that the rest of the book runs on. Nothing here is deep yet; it is the grammar we need before we can speak.

1.1 A concrete process first: the random walk

Start a token at 0. At each tick of the clock flip a fair coin and move $+1$ on heads, -1 on tails. Writing $\xi_1, \xi_2, \dots$ for the independent steps (each ± 1 with probability $\frac{1}{2}$), the position after n steps is

$$S_n = \xi_1 + \xi_2 + \dots + \xi_n, \quad S_0 = 0.$$

This **symmetric random walk** is the most important process in the book, because Brownian motion is what it becomes when the steps are made infinitely small and infinitely frequent. Two facts about it will reappear, transformed, throughout: its mean stays put, $\mathbb{E}[S_n] = 0$, while its spread grows,

$$\text{Var}(S_n) = \sum_{k=1}^n \text{Var}(\xi_k) = n,$$

so the typical distance from the origin after n steps is $\sqrt{n}$. **The displacement grows like the square root of time.** Hold onto that $\sqrt{n}$; it is the seed of $(dW)^2 = dt$.

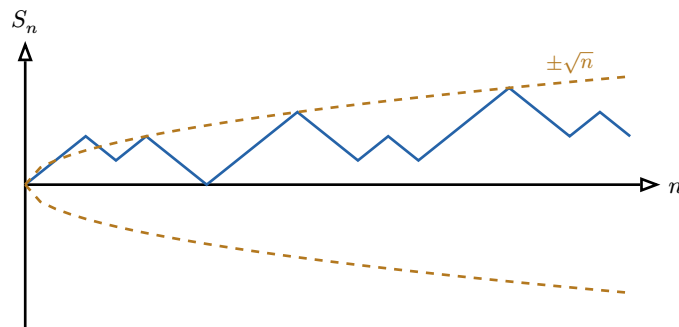


Figure 1: One sample path of the symmetric random walk (blue). The dashed curves are $\pm\sqrt{n}$: the walk wanders, but its typical magnitude tracks the square root of the number of steps, not the number of steps itself. Rescaling this picture — shrinking the steps in size while speeding them up in time — produces Brownian motion in Chapter 2.

1.2 Filtrations: information revealed over time

To speak of “what is known by time t ” we need an object that records the accumulating information. In MAT21-4 you reasoned about conditional expectation on **partitions**; the continuous-time generalization of a partition is a **σ -algebra**, and a growing family of them is a filtration.

Definition 1 (Filtration and adaptedness) A **filtration** $(\mathcal{F}_t)_{t \geq 0}$ is an increasing family of σ -algebras, $\mathcal{F}_s \subseteq \mathcal{F}_t$ whenever $s \leq t$. Think of $\mathcal{F}_t$ as **all the information available by time t** . A process (X_t) is **adapted** to the filtration if X_t is known once $\mathcal{F}_t$ is known — i.e. X_t is $\mathcal{F}_t$ -measurable — for every t .

The intuition is more important than the measure theory. $\mathcal{F}_t$ is the set of all questions that can be answered “yes/no” using only observations up to time t . The filtration **grows** because information is never forgotten: $\mathcal{F}_s \subseteq \mathcal{F}_t$. A process is adapted if it does not peek into the future — its value now depends only on what has already happened. Every process we build (the random walk, Brownian motion, the solution of an SDE) is adapted to the filtration generated by its own history, written $\mathcal{F}_t = \sigma(W_s : s \leq t)$.

Remark Adaptedness will become a **requirement**, not a convenience, when we define the Itô integral. The integrand H_t must be decided using only present information — you may bet on the next coin flip, but you may not see it first. Forbidding that peek is exactly what makes the Itô integral a fair game.

1.3 Markov chains, recalled and put in continuous-time context

A process is **Markov** if its future depends on its past only through its present: knowing where you are now, the route by which you arrived tells you nothing more about where you go next. For a discrete chain on finitely many states this is encoded in a **transition matrix** P , with $P_{ij} = \mathbb{P}(X_{n+1} = j \mid X_n = i)$, whose rows sum to 1.

Definition 2 (Stationary distribution) A probability row vector π over the states is a **stationary distribution** of the chain with transition matrix P if

$$\pi P = \pi, \quad \sum_i \pi_i = 1, \quad \pi_i \geq 0.$$

If the chain is started from π , its distribution is π at every later time: the randomness churns but the population proportions stand still.

Worked example — A two-state weather chain. Let state 1 be “sunny” and state 2 “rainy”, with

$$P = \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix}.$$

A sunny day is followed by sun with probability 0.8; a rainy day by rain with probability 0.6. To find the long-run fraction of sunny days, solve $\pi P = \pi$: writing $\pi = (a, 1 - a)$,

$$0.8a + 0.4(1 - a) = a \implies 0.4a + 0.4 = a \implies a = \frac{2}{3}.$$

So in the long run two days in three are sunny, **regardless of today’s weather**. The Markov chain forgets its starting point. The Ornstein–Uhlenbeck process of Chapter 5 is the continuous-time cousin of exactly this forgetting — its stationary distribution will be a Gaussian rather than a pair of numbers, but the equation $\pi P = \pi$ becomes the condition that the distribution no longer changes in time.

1.4 Martingales: the formal fair game

In MAT21-4 a martingale was a “fair game” — your expected fortune after the next play equals your fortune now. We can now state it with the filtration in hand.

Definition 3 (Martingale) An adapted process (M_t) with $\mathbb{E}|M_t| < \infty$ is a **martingale** with respect to $(\mathcal{F}_t)$ if for all $s \leq t$,

$$\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s.$$

The best forecast of a future value, given everything known so far, is the present value. If instead $\mathbb{E}[M_t \mid \mathcal{F}_s] \geq M_s$ the process is a **submartingale** (favourable game); with $\leq$, a **supermartingale**.

The defining equation rests on the **conditional expectation** $\mathbb{E}[\cdot \mid \mathcal{F}_s]$, which you should read exactly as in the discrete course: the best estimate of a quantity given the information $\mathcal{F}_s$. Its key property, the **tower property** $\mathbb{E}[\mathbb{E}[X \mid \mathcal{F}_t] \mid \mathcal{F}_s] = \mathbb{E}[X \mid \mathcal{F}_s]$ for $s \leq t$, is the engine behind almost every martingale computation.

Worked example — The random walk is a martingale. For the symmetric random walk S_n , with $\mathcal{F}_n$ the information from the first n flips,

$$\mathbb{E}[S_{n+1} \mid \mathcal{F}_n] = \mathbb{E}[S_n + \xi_{n+1} \mid \mathcal{F}_n] = S_n + \mathbb{E}[\xi_{n+1}] = S_n + 0 = S_n,$$

using that S_n is known given $\mathcal{F}_n$ and the next step ξ_{n+1} is independent of the past with mean 0. The walk is a martingale: a fair game in the precise sense. Brownian motion will be a martingale for the same reason, and so — crucially for pricing — will the **discounted stock price under the risk-neutral measure** in Chapter 6.

Common pitfall A martingale has **zero drift**, but zero drift is not the same as “stays near its start”. The random walk is a martingale, yet it wanders off to typical distance $\sqrt{n}$. “Fair game” constrains the **expected** increment to be zero; it says nothing about the **spread**, which can and does grow without bound.

2 Brownian motion

Brownian motion is the continuous-time limit of the random walk and the central object of the entire course. Everything we integrate, every equation we solve, is driven by it. We define it by the four properties it must satisfy, then extract the two consequences that make stochastic calculus different from ordinary calculus: its paths are continuous but **nowhere differentiable**, and its quadratic variation over $[0, t]$ equals t .

2.1 From random walk to Brownian motion

Take the symmetric random walk and rescale. Over a time interval $[0, t]$ insert n steps, each of duration $\Delta t = \frac{t}{n}$. If we kept steps of size ± 1 the variance after time t would be $n = \frac{t}{\Delta t}$, which blows up as $\Delta t \rightarrow 0$. To keep the variance finite we must shrink each step to size $\pm\sqrt{\Delta t}$. Then the variance after time t is

$$n \times (\sqrt{\Delta t})^2 = n\Delta t = t,$$

finite and independent of n . By the Central Limit Theorem of MAT31-2, the sum of these many tiny independent steps converges to a **normal** distribution with mean 0 and variance t . The limiting process is Brownian motion. Notice that the **only** step size that survives the limit is $\sqrt{\Delta t}$ — the square-root scaling is not a choice but a necessity, and it is where $(dW)^2 = dt$ is born.

2.2 The definition

Definition 4 (Brownian motion (standard Wiener process)) A process $(W_t)_{t \geq 0}$ is a **standard Brownian motion** if it satisfies the four properties:

1. **Starts at zero:** $W_0 = 0$.
2. **Independent increments:** for $0 \leq t_1 \leq t_2 \leq \dots \leq t_n$, the increments $W_{t_2} - W_{t_1}, \dots, W_{t_n} - W_{t_{n-1}}$ are mutually independent.
3. **Gaussian increments:** for $s < t$, the increment $W_t - W_s$ is normally distributed with mean 0 and variance $t - s$,

$$W_t - W_s \sim \mathcal{N}(0, t - s).$$

4. **Continuous paths:** with probability 1, the map $t \mapsto W_t$ is continuous.

Each property echoes the random walk: it starts at 0; disjoint stretches of the walk use different coin flips (independence); the rescaled increment over a span of length $t - s$ is a sum of many tiny steps, hence Gaussian with variance $t - s$ by the CLT; and shrinking the steps removes the jumps, leaving a continuous path.

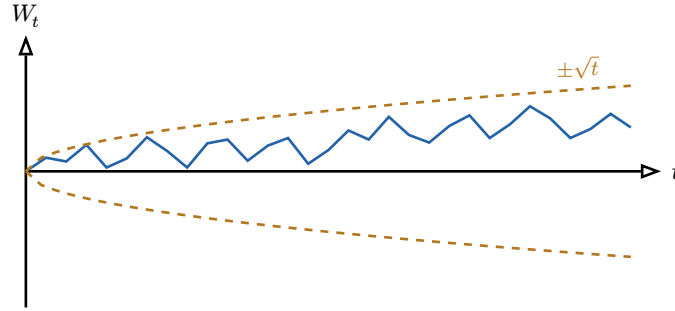


Figure 2: A simulated Brownian path (blue). It is continuous but extraordinarily rough — no matter how far you zoom in, it never straightens into a line. The dashed $\pm\sqrt{t}$ envelope marks one standard deviation: since $W_t \sim \mathcal{N}(0, t)$, the path stays within it about 68% of the time, and its spread widens like $\sqrt{t}$.

2.3 The covariance function

The increments are independent and have known variances, and that is enough to compute the covariance between the process at two different times — a quantity we will need constantly.

Proposition 5 (Covariance of Brownian motion) For all $s, t \geq 0$,

$$\text{Cov}(W_s, W_t) = \min(s, t).$$

Proof. Assume $s \leq t$ (the other case is symmetric). Split $W_t = W_s + (W_t - W_s)$. Then, since $\mathbb{E}[W_s] = \mathbb{E}[W_t] = 0$,

$$\text{Cov}(W_s, W_t) = \mathbb{E}[W_s W_t] = \mathbb{E}[W_s (W_s + (W_t - W_s))] = \mathbb{E}[W_s^2] + \mathbb{E}[W_s (W_t - W_s)].$$

The first term is $\text{Var}(W_s) = s$. In the second, $W_s = W_s - W_0$ and $W_t - W_s$ are increments over the **disjoint** intervals $[0, s]$ and $[s, t]$, hence independent with mean zero, so their product has expectation $\mathbb{E}[W_s] \mathbb{E}[W_t - W_s] = 0$. Therefore $\text{Cov}(W_s, W_t) = s = \min(s, t)$. $\square$

Remark The formula $\min(s, t)$ says that the further apart two times are, the less the earlier value predicts the later one **in relative terms**: the shared randomness is exactly the history they have in common, namely the interval $[0, \min(s, t)]$. This single function determines Brownian motion completely, because a Gaussian process is pinned down by its mean (here 0) and its covariance.

2.4 Quadratic variation: the heart of the matter

Here is the property that forces a new calculus. Chop $[0, t]$ into n equal pieces of width $\Delta t = \frac{t}{n}$ and add up the **squared** increments:

$$Q_n = \sum_{k=1}^n (W_{t_k} - W_{t_{k-1}})^2.$$

For an ordinary differentiable function f , each increment is about $f'(t_{k-1})\Delta t$, so each square is of order $(\Delta t)^2$ and the sum of n of them is of order $n(\Delta t)^2 = t\Delta t \rightarrow 0$. The squared increments of a smooth function vanish in the limit. For Brownian motion they do not.

Theorem 6 (Quadratic variation of Brownian motion) As the mesh $\Delta t \rightarrow 0$, the sum of squared increments converges (in mean square) to the length of the interval:

$$\sum_{k=1}^n (W_{t_k} - W_{t_{k-1}})^2 \rightarrow t.$$

Symbolically we write this as the multiplication rule

$$(dW)^2 = dt.$$

Proof (mean and variance of the sum). Each increment satisfies $W_{t_k} - W_{t_{k-1}} \sim \mathcal{N}(0, \Delta t)$, so $\mathbb{E}[(W_{t_k} - W_{t_{k-1}})^2] = \Delta t$, and by linearity $\mathbb{E}[Q_n] = n\Delta t = t$ for every n — the mean is **exactly** t . For the variance, use that for a $\mathcal{N}(0, \sigma^2)$ variable Z one has $\text{Var}(Z^2) = 2\sigma^4$. The squared increments are independent, so

$$\text{Var}(Q_n) = \sum_{k=1}^n 2(\Delta t)^2 = 2n(\Delta t)^2 = 2t\Delta t.$$

As $\Delta t \rightarrow 0$ this tends to 0. A random variable whose mean is t and whose variance vanishes converges to the constant t . $\square$

The contrast could not be sharper, and it is worth a figure.

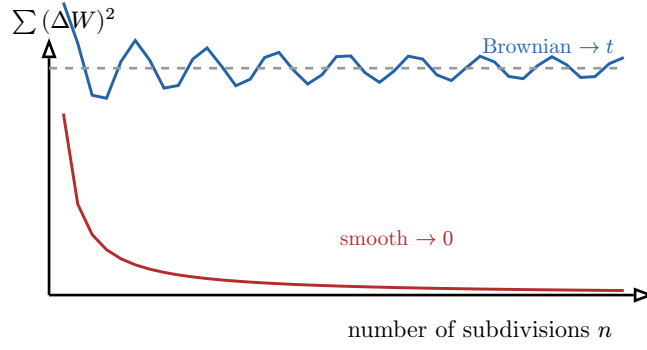


Figure 3: Sum of squared increments as the partition is refined. For a smooth function (red) the sum collapses to zero — this is why classical calculus can discard $(dx)^2$. For Brownian motion (blue) it settles on the interval length t and refuses to vanish. That stubborn, non-zero quadratic variation is the entire reason stochastic calculus exists.

2.5 Why Brownian paths are nowhere differentiable

Quadratic variation immediately explains the most startling property of Brownian motion. If W were differentiable at some time t , then near t we would have $W_{t+\Delta t} - W_t \approx W'_t \Delta t$, an increment of order Δt . But the **actual** typical increment is $\sqrt{\Delta t}$ — vastly larger than Δt when Δt is small. The ratio

$$\frac{W_{t+\Delta t} - W_t}{\Delta t} \sim \frac{\sqrt{\Delta t}}{\Delta t} = \frac{1}{\sqrt{\Delta t}} \rightarrow \infty$$

does not settle on any finite slope; it explodes. A Brownian path is continuous (no jumps) yet so jagged that it has a well-defined direction at no single instant.

Common pitfall This is why you must never write “ dW/dt ”. There is no such derivative. The notation dW_t is **not** a velocity times dt ; it is an honest infinitesimal increment, a $\mathcal{N}(0, dt)$ random variable, meaningful only inside an integral $\int H_t dW_t$. Treating dW as if it were a smooth $W'dt$ is the single most common and most fatal error in the subject.

3 Itô integration

Because Brownian paths are not differentiable, the integral $\int_0^t H_s dW_s$ cannot be defined path-by-path as a Riemann–Stieltjes integral — the machinery of MAT31-2 simply does not apply. Itô’s insight was to define the integral as a **limit in mean square** of sums that evaluate the integrand at the **left** endpoint of each subinterval. That single choice — left endpoint, decided before the increment is revealed — is what makes the integral a martingale and gives it its two signature properties.

3.1 Building the integral for simple integrands

We start, as Itô did, with the simplest possible integrands: those that are piecewise constant in time.

Definition 7 (Simple integrand) A process (H_t) is **simple** if there is a partition $0 = t_0 < t_1 < \dots < t_n = T$ such that $H_t = H_{t_{k-1}}$ is constant on each interval $[t_{k-1}, t_k)$, and each $H_{t_{k-1}}$ is $\mathcal{F}_{t_{k-1}}$ -measurable — **known at the start of the interval**.

For such an integrand the Itô integral is defined by the only formula that could possibly be right: value times increment, summed.

Definition 8 (Itô integral of a simple integrand)

$$\int_0^T H_s dW_s := \sum_{k=1}^n H_{t_{k-1}} (W_{t_k} - W_{t_{k-1}}).$$

The integrand on $[t_{k-1}, t_k)$ multiplies the Brownian increment over that same interval, and the multiplier is locked in at the **left** endpoint t_{k-1} .

The general integral is then obtained by approximating any adapted, square-integrable integrand by simple ones and passing to the mean-square limit; the isometry below is precisely what guarantees the limit exists and is unique. The financial reading makes the left-endpoint rule inevitable: $H_{t_{k-1}}$ is the number of shares you hold **going into** the period $[t_{k-1}, t_k)$, and $W_{t_k} - W_{t_{k-1}}$ is the price shock that **then** arrives. You must choose your position before the shock, not after — anything else is insider trading, not investing.

Common pitfall The left-endpoint choice is not a harmless convention as it is in ordinary calculus. For a smooth integrand the Riemann sum converges to the same limit whether you evaluate at the left, right, or midpoint. For Itô integration the endpoint **changes the answer**, precisely because $(dW)^2 = dt$ does not vanish. Evaluating at the midpoint instead gives a different object, the **Stratonovich integral**, which obeys the ordinary chain rule but is not a martingale. Throughout this course “stochastic integral” means **Itô**: left endpoint, always.

3.2 The zero-expectation property

Proposition 9 (Zero expectation) For an adapted, square-integrable integrand H ,

$$\mathbb{E} \left[\int_0^T H_s dW_s \right] = 0.$$

Proof. It suffices to check it for a simple integrand. Take expectations termwise:

$$\mathbb{E} \left[\sum_k H_{t_{k-1}} (W_{t_k} - W_{t_{k-1}}) \right] = \sum_k \mathbb{E} [H_{t_{k-1}} (W_{t_k} - W_{t_{k-1}})].$$

Condition each term on $\mathcal{F}_{t_{k-1}}$. Since $H_{t_{k-1}}$ is known given that information, it pulls out, and the future increment has conditional mean zero:

$$\mathbb{E} [H_{t_{k-1}} (W_{t_k} - W_{t_{k-1}})] = \mathbb{E} \left[H_{t_{k-1}} \underbrace{\mathbb{E} [W_{t_k} - W_{t_{k-1}} | \mathcal{F}_{t_{k-1}}]}_{=0} \right] = 0.$$

Every term vanishes, so the whole sum does. □

This is the continuous-time statement that **betting a fair game with a strategy fixed in advance is still fair**. It is the reason the Itô integral, viewed as a process in its upper limit

T , is itself a martingale — and the reason risk-neutral pricing in Chapter 6 reduces to “take an expectation”.

3.3 The Itô isometry

The zero-expectation property tells us the integral’s mean; the **Itô isometry** tells us its variance, and it is the workhorse of every computation in the subject.

Theorem 10 (Itô isometry) For an adapted, square-integrable integrand H ,

$$\mathbb{E} \left[\left(\int_0^T H_s dW_s \right)^2 \right] = \mathbb{E} \left[\int_0^T H_s^2 ds \right].$$

The mean square of a stochastic integral equals the ordinary (time) integral of the mean square of the integrand.

Proof (simple integrand). Write $\Delta W_k = W_{t_k} - W_{t_{k-1}}$. Squaring the defining sum,

$$\left(\sum_k H_{t_{k-1}} \Delta W_k \right)^2 = \sum_k H_{t_{k-1}}^2 (\Delta W_k)^2 + 2 \sum_{j < k} H_{t_{j-1}} H_{t_{k-1}} \Delta W_j \Delta W_k.$$

Take expectations. For a **cross** term $j < k$, condition on $\mathcal{F}_{t_{k-1}}$: everything except the last increment ΔW_k is known, and $\mathbb{E}[\Delta W_k | \mathcal{F}_{t_{k-1}}] = 0$, so every cross term vanishes. For a **diagonal** term, $H_{t_{k-1}}^2$ is independent of the future increment and $\mathbb{E}[(\Delta W_k)^2] = t_k - t_{k-1}$, giving

$$\mathbb{E}[H_{t_{k-1}}^2 (\Delta W_k)^2] = \mathbb{E}[H_{t_{k-1}}^2] (t_k - t_{k-1}).$$

Summing, $\mathbb{E}[(\int H dW)^2] = \sum_k \mathbb{E}[H_{t_{k-1}}^2] (t_k - t_{k-1}) = \mathbb{E}[\int_0^T H_s^2 ds]$, the last step recognising the Riemann sum of H_s^2 . $\square$

The isometry is exactly $(dW)^2 = dt$ wearing the clothes of an expectation: the squared increment $(\Delta W_k)^2$ became $t_k - t_{k-1}$, turning a sum of **stochastic** squares into a sum of plain time-lengths. This is why a slogan suffices to remember it: **to square a dW integral, replace dW by ds and drop the cross terms.**

Worked example — Variance of a stochastic integral. Compute $\text{Var}(\int_0^T s dW_s)$. By zero expectation the mean is 0, so the variance equals the second moment, and the isometry gives

$$\mathbb{E} \left[\left(\int_0^T s dW_s \right)^2 \right] = \mathbb{E} \left[\int_0^T s^2 ds \right] = \int_0^T s^2 ds = \frac{T^3}{3}.$$

No probability is left in the final computation — the isometry has converted a hard stochastic question into a freshman integral.

Worked example — The integral of Brownian motion against itself. A famous calculation that previews Itô’s formula: we claim

$$\int_0^T W_s dW_s = \frac{1}{2} W_T^2 - \frac{1}{2} T.$$

Naïve calculus would predict just $\frac{1}{2}W_T^2$, by analogy with $\int x dx = \frac{1}{2}x^2$. The extra $-\frac{1}{2}T$ is the quadratic variation making itself felt. To see it, telescope $W_T^2 = \sum_k (W_{t_k}^2 - W_{t_{k-1}}^2)$ and use the algebraic identity $b^2 - a^2 = 2a(b - a) + (b - a)^2$ with $a = W_{t_{k-1}}$, $b = W_{t_k}$:

$$W_T^2 = 2 \sum_k W_{t_{k-1}} \Delta W_k + \sum_k (\Delta W_k)^2.$$

The first sum is $2 \int_0^T W_s dW_s$ (left endpoints!) and the second converges to T by quadratic variation. Rearranging gives the claim. The missing $-\frac{1}{2}T$ is the signature of stochastic calculus, and the next chapter turns this one-off trick into a universal rule.

4 Itô's formula

Itô's formula is the chain rule of stochastic calculus, and it is the single most useful result in the book. It tells us how a smooth function of a stochastic process evolves — and it carries an extra term, absent from ordinary calculus, that comes from exactly one place: $(dW)^2 = dt$.

4.1 The need for a corrected chain rule

We already have a warning shot. The previous chapter showed $\int_0^T W_s dW_s = \frac{1}{2}W_T^2 - \frac{1}{2}T$, equivalently in differential form

$$d(W_t^2) = 2W_t dW_t + dt.$$

Ordinary calculus, applying the chain rule to $f(W) = W^2$, would give only $d(W^2) = 2W dW$. The surplus $+dt$ is the correction. Itô's formula is the general law that produces it for **any** smooth f , and the mechanism is a Taylor expansion in which the second-order term survives.

4.2 Itô's formula for a function of Brownian motion

Theorem 11 (Itô's formula, basic form) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be twice continuously differentiable (C^2). Then

$$df(W_t) = f'(W_t)dW_t + \frac{1}{2}f''(W_t)dt,$$

or in integral form

$$f(W_t) = f(W_0) + \int_0^t f'(W_s)dW_s + \frac{1}{2} \int_0^t f''(W_s)ds.$$

Proof (Taylor expansion heuristic). Expand f to second order over a small step:

$$f(W_{t+dt}) - f(W_t) = f'(W_t)dW + \frac{1}{2}f''(W_t)(dW)^2 + \text{higher order}.$$

In ordinary calculus the $(dW)^2$ term is discarded as negligible. Here it is **not** negligible: by quadratic variation $(dW)^2 = dt$, so it contributes $\frac{1}{2}f''(W_t)dt$. Terms of order $(dt)^2$ and $dW \cdot dt$ are genuinely negligible and drop out. Summing the increments gives the integral form. $\square$

The whole novelty lives in the highlighted half-second-derivative term.

The Itô multiplication table. Every stochastic Taylor expansion is governed by three rules, applied mechanically:

$$(dW)^2 = dt, \quad dW \cdot dt = 0, \quad (dt)^2 = 0.$$

Keep terms up to order dt ; everything smaller vanishes. The first rule is the new physics; the other two say “anything smaller than dt is nothing”.

Common pitfall The function must be C^2 for the formula to hold — the second derivative f'' has to exist and be continuous, because the correction term needs it. This is why $f(x) = |x|$ or the option payoff $\max(x - K, 0)$ cannot be fed directly into Itô’s formula at their kinks; one must smooth them or treat the kink separately.

4.3 The general Itô process and its formula

In practice the process driving f is rarely W itself but an **Itô process** with its own drift and diffusion.

Definition 12 (Itô process) A process (X_t) is an **Itô process** if it satisfies

$$dX_t = \mu_t dt + \sigma_t dW_t,$$

where μ_t (the **drift**) and σ_t (the **diffusion** or **volatility**) are adapted. The drift is the predictable, average rate of change; the diffusion scales the random shocks.

Applying the Itô multiplication table to a C^2 function of such a process — and remembering $(dX)^2 = \sigma_t^2 (dW)^2 = \sigma_t^2 dt$ — gives the form we use constantly.

Theorem 13 (Itô’s formula for an Itô process) If $dX_t = \mu_t dt + \sigma_t dW_t$ and f is C^2 (or $C^{1,2}$ in (t, x) if f also depends on time), then

$$df(t, X_t) = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial x} dX_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2} \sigma_t^2 dt.$$

Grouping into drift and diffusion,

$$df = \underbrace{\left(f_t + \mu_t f_x + \frac{1}{2} \sigma_t^2 f_{xx} \right)}_{\text{drift}} dt + \underbrace{\sigma_t f_x}_{\text{diffusion}} dW_t.$$

Worked example — Geometric Brownian motion via Itô. Let $X_t = \exp(\sigma W_t + (\mu - \frac{1}{2}\sigma^2)t)$, the process we will use for a stock price. Take $f(t, x) = \exp(\sigma x + (\mu - \frac{1}{2}\sigma^2)t)$ with $x = W_t$. Then $f_t = (\mu - \frac{1}{2}\sigma^2)f$, $f_x = \sigma f$, $f_{xx} = \sigma^2 f$, and since the driving process is W ($\mu_t = 0, \sigma_t = 1$),

$$dX_t = \left[\left(\mu - \frac{1}{2}\sigma^2 \right) f + \frac{1}{2}\sigma^2 f \right] dt + \sigma f dW_t = \mu X_t dt + \sigma X_t dW_t.$$

The two $\frac{1}{2}\sigma^2$ terms cancel, leaving $dX_t = \mu X_t dt + \sigma X_t dW_t$. So this explicit exponential **is** the solution of the geometric-Brownian-motion SDE — a fact we will derive from scratch in the next chapter, here verified in one line by Itô.

4.4 Computing expectations with Itô

The drift–diffusion split is not just bookkeeping; it is a computational tool. The diffusion part $\int \sigma f_x dW$ is a martingale, so by the zero- expectation property it **disappears under expectation**. Whatever is left — the drift — governs how the mean evolves.

Worked example — The mean of a transformed process. What is $\mathbb{E}[W_t^4]$? Apply Itô to $f(x) = x^4$: $f' = 4x^3$, $f'' = 12x^2$, so

$$d(W_t^4) = 4W_t^3 dW_t + \frac{1}{2} \cdot 12W_t^2 dt = 4W_t^3 dW_t + 6W_t^2 dt.$$

Take expectations; the dW term has mean zero, leaving the **ordinary** ODE

$$\frac{d}{dt} \mathbb{E}[W_t^4] = 6\mathbb{E}[W_t^2] = 6t,$$

using $\mathbb{E}[W_t^2] = t$. Integrating with $\mathbb{E}[W_0^4] = 0$ gives $\mathbb{E}[W_t^4] = 3t^2$. We have recovered the fourth moment of a $\mathcal{N}(0, t)$ variable with no integration against a density at all — Itô turned a moment computation into the MAT41-4 exercise of solving a trivial ODE.

Remark This is the template for Learning Outcome 4 and a pattern you will use everywhere: write df by Itô, take expectations to kill the martingale part, and read off an ordinary differential equation for the mean. Stochastic calculus hands its hardest questions back to the deterministic calculus you already know.

5 Stochastic differential equations

A **stochastic differential equation** (SDE) is the noisy analogue of the ODEs of MAT41-4: a law of motion in which, alongside the deterministic drift, a Brownian term injects randomness at every instant. This chapter states when an SDE has a unique solution and solves the three models that anchor the rest of the course: geometric Brownian motion, the Ornstein–Uhlenbeck process, and the CIR model.

5.1 Drift, diffusion, and what a solution means

Definition 14 (Stochastic differential equation) An SDE is an equation of the form

$$dX_t = \mu(t, X_t)dt + \sigma(t, X_t)dW_t, \quad X_0 = x_0,$$

with **drift coefficient** $\mu(t, x)$ and **diffusion coefficient** $\sigma(t, x)$. A **solution** is an adapted process (X_t) satisfying the integral equation

$$X_t = x_0 + \int_0^t \mu(s, X_s)ds + \int_0^t \sigma(s, X_s)dW_s,$$

in which the second integral is an Itô integral.

Set $\sigma \equiv 0$ and you recover exactly the Cauchy problem $X' = \mu(t, X)$, $X(0) = x_0$ of MAT41-4. The diffusion term is the new ingredient, and the “derivative” form is only shorthand for the integral equation, since X has no genuine derivative.

5.2 Existence and uniqueness

The Cauchy–Lipschitz theorem of MAT41-4 guaranteed a unique ODE solution when the right-hand side was Lipschitz. The SDE version is almost identical in spirit, with the same two conditions now imposed on **both** coefficients.

Theorem 15 (Existence and uniqueness for SDEs) Suppose the coefficients satisfy, for some constant K and all x, y, t :

- **Lipschitz continuity:**

$$|\mu(t, x) - \mu(t, y)| + |\sigma(t, x) - \sigma(t, y)| \leq K |x - y|;$$

- **Linear growth:**

$$|\mu(t, x)| + |\sigma(t, x)| \leq K(1 + |x|).$$

Then the SDE has a unique (strong) solution (X_t) on $[0, T]$ with continuous paths.

The two conditions play the same roles as in the deterministic theory. **Lipschitz continuity** stops two solutions starting nearby from separating too fast, forcing uniqueness — its failure is what allowed the ODE $y' = \sqrt{|y|}$ to have many solutions in MAT41-4. **Linear growth** stops the coefficients from exploding, so the solution cannot run off to infinity in finite time.

Worked example — Checking the conditions for GBM. For geometric Brownian motion $\mu(t, x) = \mu x$ and $\sigma(t, x) = \sigma x$ are both linear in x . Linearity is Lipschitz ($|\mu x - \mu y| = |\mu| |x - y|$) and has linear growth, so a unique solution exists. Contrast the CIR diffusion $\sigma\sqrt{x}$ below, whose square root is **not** Lipschitz at $x = 0$ — a genuine subtlety we will have to flag.

5.3 Geometric Brownian motion

GBM is the workhorse model for a quantity that grows at a **percentage** rate with **percentage** shocks — a stock price, a population, an investment.

Definition 16 (Geometric Brownian motion) GBM is the solution of

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad S_0 > 0,$$

where μ is the **drift rate** (expected return) and σ the **volatility**. Both the drift and the noise scale with the current level S_t .

We solve it by the **log substitution**, the stochastic echo of separating variables. The trick is to apply Itô to $Y_t = \ln S_t$ — and the celebrated $-\frac{1}{2}\sigma^2$ term appears precisely because $\ln$ is nonlinear.

Worked example — Solving GBM by the log substitution. Let $f(x) = \ln x$, so $f'(x) = \frac{1}{x}$ and $f''(x) = -\frac{1}{x^2}$. With $dS_t = \mu S_t dt + \sigma S_t dW_t$ and diffusion $\sigma_t = \sigma S_t$, Itô's formula gives

$$d(\ln S_t) = \frac{1}{S_t} dS_t + \frac{1}{2} \left(-\frac{1}{S_t^2} \right) (\sigma S_t)^2 dt = (\mu dt + \sigma dW_t) - \frac{1}{2} \sigma^2 dt = \left(\mu - \frac{1}{2} \sigma^2 \right) dt + \sigma dW_t.$$

The right-hand side has **constant** coefficients, so $\ln S_t$ is just Brownian motion with drift and integrates directly:

$$\ln S_t = \ln S_0 + \left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma W_t.$$

Exponentiating,

$$S_t = S_0 \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W_t\right).$$

Two consequences deserve emphasis. First, $S_t > 0$ always — the exponential can never reach zero — which is exactly why GBM models prices, that cannot go negative. Second, taking expectations of the lognormal,

$$\mathbb{E}[S_t] = S_0 e^{\mu t},$$

so the **expected** price grows at the clean rate μ , even though the **typical** (median) log-price drifts at the slower $\mu - \frac{1}{2}\sigma^2$.

Common pitfall The gap between μ (the growth rate of the **mean**) and $\mu - \frac{1}{2}\sigma^2$ (the growth rate of the **median**, the log) is not a mistake — it is **volatility drag**. Because the exponential is convex, the same volatility that scatters outcomes pulls the median below the mean. Confusing these two rates is a classic and expensive error; the $-\frac{1}{2}\sigma^2$ is real money.

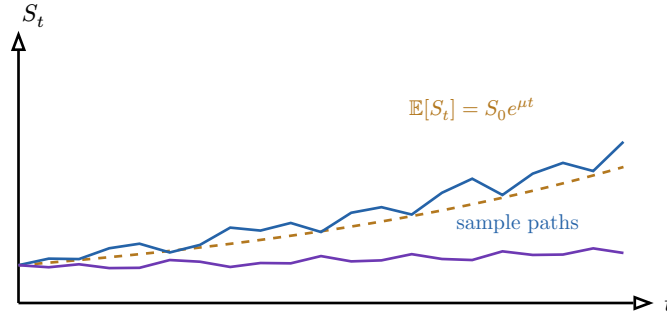


Figure 4: Two simulated GBM price paths (blue, purple) around the expected-value curve $\mathbb{E}[S_t] = S_0 e^{\mu t}$ (dashed). The paths stay strictly positive because S_t is an exponential, and they fan out multiplicatively — larger prices wobble by larger absolute amounts. This is the standard model of a stock price in the Black–Scholes world of the next chapter.

5.4 The Ornstein–Uhlenbeck process and mean reversion

Not everything wanders off like GBM. Interest rates, volatilities, and temperatures are **mean-reverting**: pulled back toward a long-run level whenever they stray. The Ornstein–Uhlenbeck (OU) process is the canonical model.

Definition 17 (Ornstein–Uhlenbeck process)

$$dX_t = \theta(m - X_t)dt + \sigma dW_t,$$

with **reversion speed** $\theta > 0$, **long-run mean** m , and constant volatility σ . The drift $\theta(m - X_t)$ is a restoring force: positive when $X_t < m$ (push up), negative when $X_t > m$ (push down).

The drift is linear, so we can solve it explicitly with an **integrating factor**, exactly the MAT41-4 method for first-order linear ODEs, now applied through Itô.

Worked example — Solving OU by the integrating factor. Multiply through by $e^{\theta t}$ and apply Itô to $Y_t = e^{\theta t} X_t$ (here $f(t, x) = e^{\theta t} x$, so $f_t = \theta e^{\theta t} x$, $f_x = e^{\theta t}$, $f_{xx} = 0$):

$$d(e^{\theta t} X_t) = \theta e^{\theta t} X_t dt + e^{\theta t} dX_t = \theta e^{\theta t} X_t dt + e^{\theta t} [\theta(m - X_t) dt + \sigma dW_t].$$

The X_t terms cancel, leaving

$$d(e^{\theta t} X_t) = \theta m e^{\theta t} dt + \sigma e^{\theta t} dW_t.$$

Integrating from 0 to t and dividing by $e^{\theta t}$,

$$X_t = m + (X_0 - m)e^{-\theta t} + \sigma \int_0^t e^{-\theta(t-s)} dW_s.$$

Read the solution in three pieces: the long-run level m ; a **deterministic relaxation** $(X_0 - m)e^{-\theta t}$ that decays the initial gap at rate θ ; and a stochastic integral of past shocks, **exponentially discounted** so the distant past is forgotten. From it we read off the moments. The mean reverts, $\mathbb{E}[X_t] = m + (X_0 - m)e^{-\theta t} \rightarrow m$, and the variance, by the Itô isometry,

$$\text{Var}(X_t) = \sigma^2 \int_0^t e^{-2\theta(t-s)} ds = \frac{\sigma^2}{2\theta} (1 - e^{-2\theta t}) \rightarrow \frac{\sigma^2}{2\theta}.$$

Proposition 18 (Stationary distribution of OU) As $t \rightarrow \infty$ the OU process converges to its **stationary distribution**

$$X_\infty \sim \mathcal{N}\left(m, \frac{\sigma^2}{2\theta}\right).$$

The mean settles at the long-run level m ; the stationary variance balances the noise input σ^2 against the restoring pull 2θ .

This Gaussian is the continuous-time analogue of the stationary vector π of Chapter 1: the distribution that no longer changes once the process has run long enough. Faster reversion (larger θ) or weaker noise (smaller σ) yields a tighter stationary spread.

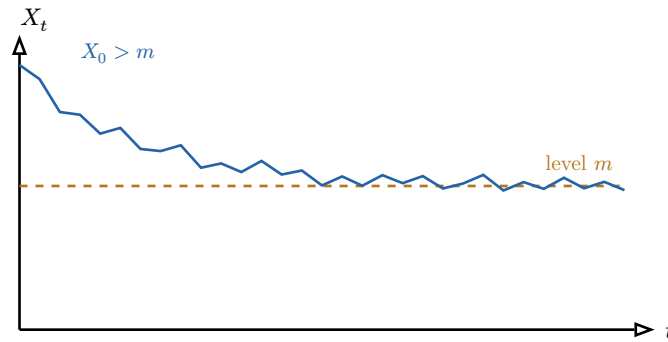


Figure 5: An Ornstein–Uhlenbeck path started above its long-run level m (dashed). The restoring drift $\theta(m - X_t)$ continually pulls it back, while the noise σdW keeps it jittering; the result hovers around m rather than wandering off as Brownian motion or GBM would. In the long run its law settles to the Gaussian $\mathcal{N}\left(m, \frac{\sigma^2}{2\theta}\right)$.

5.5 The CIR model

A drawback of OU as an interest-rate model is that its Gaussian law allows **negative** values. The Cox–Ingersoll–Ross (CIR) model fixes this with a square-root diffusion.

Definition 19 (CIR process)

$$dX_t = \theta(m - X_t)dt + \sigma\sqrt{X_t}dW_t, \quad X_0 > 0.$$

It keeps the mean-reverting drift of OU but multiplies the noise by $\sqrt{X_t}$, so the volatility shrinks as X_t approaches 0.

The $\sqrt{X_t}$ factor is the whole idea: as the process nears zero the noise fades, and the upward drift θm dominates and lifts it back, so X_t stays non-negative. If the **Feller condition** $2\theta m \geq \sigma^2$ holds, the process never even touches zero. The price is theoretical delicacy — $\sqrt{x}$ is not Lipschitz at 0, so the clean existence theorem above does not directly apply, and a more careful argument (using the specific structure) is needed. This is the same Lipschitz-failure-at-a-square-root phenomenon you met in MAT41-4 with $y' = \sqrt{|y|}$, here tamed by the drift rather than breaking uniqueness.

6 The Black–Scholes model

Everything so far has been preparation for this chapter — the reason stochastic calculus earns its place in a quantitative-finance curriculum. We model a stock as geometric Brownian motion, combine Itô’s formula with a **hedging** argument to derive the Black–Scholes partial differential equation, value options under the **risk-neutral** measure, and read off the price of a European call together with its sensitivities, the Greeks.

6.1 The setup: stock, bond, and option

Two instruments are available. A **risk-free bond** grows deterministically at the continuously compounded rate r ,

$$dB_t = rB_t dt, \quad B_t = B_0 e^{rt},$$

which is just the continuous-compounding ODE of MAT41-4. A **stock** follows GBM under the real-world measure,

$$dS_t = \mu S_t dt + \sigma S_t dW_t.$$

An **option** is a contract whose value $V(t, S_t)$ depends on the stock; a **European call** with strike K and maturity T pays, at time T ,

$$V(T, S_T) = \max(S_T - K, 0) =: (S_T - K)^+,$$

the right — but not the obligation — to buy the stock for K . Our task is to find its fair value $V(t, S)$ at all earlier times.

6.2 Deriving the Black–Scholes PDE by delta hedging

The decisive idea is **hedging**: build a portfolio of stock and option whose random shocks exactly cancel, leaving a riskless position. Since it bears no risk, it must earn the risk-free rate r — otherwise arbitrage. Forcing that gives a deterministic equation for V .

Theorem 20 (Black–Scholes PDE) Under the GBM model, any option value $V(t, S)$ written on the stock satisfies

$$\frac{\partial V}{\partial t} + rS \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rV = 0,$$

with the option's payoff as terminal condition, e.g. $V(T, S) = (S - K)^+$ for a call.

Proof (delta hedging). First, by Itô's formula applied to $V(t, S_t)$ with $dS_t = \mu S_t dt + \sigma S_t dW_t$ (so the diffusion is σS_t and $(dS)^2 = \sigma^2 S_t^2 dt$),

$$dV = \left(V_t + \mu S_t V_S + \frac{1}{2} \sigma^2 S_t^2 V_{SS} \right) dt + \sigma S_t V_S dW_t.$$

Now form a portfolio: hold one option and **short** $\Delta = V_S$ shares, $\Pi_t = V - V_S S_t$. Its change is

$$d\Pi = dV - V_S dS_t = \left(V_t + \frac{1}{2} \sigma^2 S_t^2 V_{SS} \right) dt,$$

where the two $\sigma S_t V_S dW_t$ terms — one from dV , one from $V_S dS$ — **cancel exactly**. The portfolio is now riskless: no dW remains. A riskless portfolio must earn r : $d\Pi = r\Pi dt = r(V - V_S S_t) dt$. Equating the two expressions for $d\Pi$ and cancelling dt ,

$$V_t + \frac{1}{2} \sigma^2 S_t^2 V_{SS} = rV - rS_t V_S,$$

which rearranges to the stated PDE. □

Remark (Where the drift went) Look closely at the PDE: the stock's real-world drift μ has **vanished**. The hedge removed not only the randomness but also the expected return — the option's value cannot depend on whether you personally think the stock will rise, because the hedger neutralises that view. This disappearance of μ is the mathematical seed of risk-neutral pricing.

6.3 Risk-neutral valuation

The PDE has a probabilistic solution that is far easier to use than the equation itself. Because μ dropped out, we may **pretend** the stock drifts at the risk-free rate r instead of μ — this defines the **risk-neutral measure** $\mathbb{Q}$, under which

$$dS_t = rS_t dt + \sigma S_t dW_t^{\mathbb{Q}}.$$

Theorem 21 (Risk-neutral pricing formula) The fair value of an option with payoff $\Phi(S_T)$ at maturity T is its **discounted expected payoff under $\mathbb{Q}$** :

$$V(t, S_t) = e^{-r(T-t)} \mathbb{E}^{\mathbb{Q}}[\Phi(S_T) \mid S_t].$$

This is the single most useful formula in quantitative finance, and it reduces pricing to the Monte Carlo integration of MAT31-2 — simulate S_T under $\mathbb{Q}$, average the payoff, discount. The transformation “ $\mu \rightarrow r$ ” is the entire content of the change of measure: drift at the risk-free rate, then take an expectation.

Common pitfall The risk-neutral measure is a **pricing device**, not a prediction. Stocks do not actually drift at r in the real world — they drift at $\mu > r$ to compensate for risk. The

probabilities under $\mathbb{Q}$ are not the true probabilities of outcomes; they are the weights that make every traded asset earn r on average, which is exactly what no-arbitrage demands. Never report a risk-neutral expected price as a real-world forecast.

6.4 The Black–Scholes formula

For a European call the risk-neutral expectation can be computed in closed form, because S_T is lognormal under $\mathbb{Q}$. The result is the most famous formula in finance.

Theorem 22 (Black–Scholes formula for a European call) With time to maturity $\tau = T - t$, the price of a European call is

$$C(t, S) = S\Phi(d_1) - Ke^{-r\tau}\Phi(d_2),$$

where Φ is the standard normal CDF and

$$d_1 = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \quad d_2 = d_1 - \sigma\sqrt{\tau}.$$

The two terms have clean meanings. $S\Phi(d_1)$ is the present value of receiving the stock if the option finishes in the money; $Ke^{-r\tau}\Phi(d_2)$ is the present value of paying the strike K , weighted by $\Phi(d_2) = \mathbb{Q}(S_T > K)$, the risk-neutral probability of exercise. The factor $e^{-r\tau}$ is the same discounting you have used since MAT31-2.

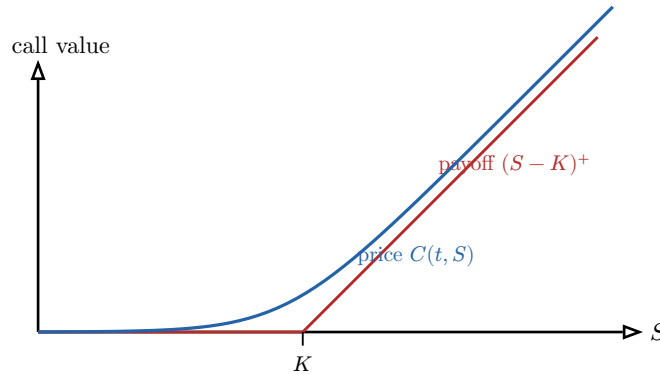


Figure 6: The Black–Scholes call price $C(t, S)$ (blue) as a function of the current stock price, sitting above the maturity payoff $(S - K)^+$ (red kinked line). Before expiry the curve is smooth and strictly above the payoff: the gap is **time value**, the worth of the remaining optionality. As $t \rightarrow T$ the smooth curve relaxes down onto the kinked payoff.

6.5 The Greeks

The **Greeks** are the partial derivatives of the option price — its sensitivities to the variables it depends on. They are what a trader actually hedges, and they follow by differentiating the Black–Scholes formula.

Definition 23 (The principal Greeks) For the call price $C(t, S)$:

- **Delta** $\Delta = \frac{\partial C}{\partial S} = \Phi(d_1)$ — sensitivity to the stock price; the number of shares to hold to hedge.
- **Gamma** $\Gamma = \frac{\partial^2 C}{\partial S^2} = \frac{\varphi(d_1)}{S\sigma\sqrt{\tau}}$ — sensitivity of Delta itself, the curvature of the price; here φ is the normal density.

- **Vega** $\mathcal{V} = \frac{\partial C}{\partial \sigma} = S\varphi(d_1)\sqrt{\tau}$ — sensitivity to volatility.

Delta is exactly the hedge ratio V_S that appeared in the PDE derivation: holding Δ shares against the option neutralises first-order stock moves. Gamma measures how fast that hedge goes stale as the stock moves — high Gamma means you must re-hedge often. Vega quantifies exposure to volatility, the one input that cannot be observed directly and must be estimated.

Common pitfall Delta-hedging is only **instantaneously** riskless. Because Gamma is non-zero, the moment the stock moves, $\Delta = \Phi(d_1)$ changes and the hedge is no longer perfect; the derivation assumed **continuous** re-hedging. In practice one re-hedges at discrete times and carries residual Gamma risk — the gap between the idealised model and tradeable reality.

6.6 Put–call parity

A model-independent relationship ties the call and put prices together, and it follows from a pure no-arbitrage argument — no PDE required.

Proposition 24 (Put–call parity) For European options with the same strike K and maturity T ,

$$C - P = S - Ke^{-r(T-t)},$$

where C and P are the call and put prices.

Proof (replication). Consider two portfolios. Portfolio A: one call plus cash $Ke^{-r(T-t)}$. Portfolio B: one put plus one share. At maturity, A is worth $(S_T - K)^+ + K$ and B is worth $(K - S_T)^+ + S_T$; a quick check of the cases $S_T \geq K$ and $S_T < K$ shows both equal $\max(S_T, K)$. Equal payoffs in every state force equal prices today (else arbitrage): $C + Ke^{-r(T-t)} = P + S$, which rearranges to the claim. $\square$

Remark Put–call parity is a sanity check you can apply without trusting any model: it holds whatever the true dynamics of S , because it rests only on replication and no-arbitrage. If quoted call and put prices ever violate it, there is a free lunch — and in liquid markets, there is not.

7 Monte Carlo option pricing

When the payoff is too complicated for a closed-form risk-neutral expectation — path-dependent options, baskets, early features — we fall back on simulation. This is the Monte Carlo method of MAT31-2, now applied to the SDE-driven price: discretize the dynamics, simulate many risk-neutral paths, average the discounted payoff, and report a confidence interval. The final chapter closes the loop from abstract Itô calculus back to runnable computation.

7.1 Euler–Maruyama discretization

To simulate S_t we cannot integrate the SDE exactly in general, so we step it forward on a time grid, exactly as Euler’s method stepped an ODE in MAT41-4 — with one crucial change forced by Brownian motion.

Definition 25 (Euler–Maruyama scheme) Partition $[0, T]$ into steps of size Δt . For $dX_t = \mu(X_t)dt + \sigma(X_t)dW_t$, update

$$X_{k+1} = X_k + \mu(X_k)\Delta t + \sigma(X_k)\sqrt{\Delta t}Z_k,$$

where $Z_k \sim \mathcal{N}(0, 1)$ are independent standard normals.

The drift term $\mu(X_k)\Delta t$ is exactly Euler’s method. The novelty is the diffusion term, which scales as $\sqrt{\Delta t}$, **not** Δt — because the Brownian increment over a step of length Δt has standard deviation $\sqrt{\Delta t}$. Generating $\Delta W_k = \sqrt{\Delta t}Z_k$ reproduces exactly $\mathcal{N}(0, \Delta t)$. This $\sqrt{\Delta t}$ scaling is $(dW)^2 = dt$ appearing one last time, now as a line of code.

Common pitfall The most common simulation bug is scaling the noise by Δt instead of $\sqrt{\Delta t}$. With $\Delta t = 0.01$ the two differ by a factor of ten, so the mistake silently **throttles all the volatility** out of the model — paths look far too smooth and every price comes out wrong. Whenever a simulated price is suspiciously close to the deterministic one, check this scaling first.

Worked example — Simulating a stock under $\mathbb{Q}$. To price under the risk-neutral measure we simulate GBM with drift r :

$$S_{k+1} = S_k + rS_k\Delta t + \sigma S_k\sqrt{\Delta t}Z_k.$$

For GBM we can do better and avoid discretization error entirely, by stepping the **exact** log-solution,

$$S_{k+1} = S_k \exp\left(\left(r - \frac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t}Z_k\right),$$

which is exact at the grid points because we know GBM’s closed form. For models without a closed form (CIR, most exotics) the plain Euler–Maruyama step is the general tool.

7.2 The Monte Carlo estimator and its confidence interval

Having simulated N independent terminal prices $S_T^{(1)}, \dots, S_T^{(N)}$ under $\mathbb{Q}$, we estimate the price by the discounted sample mean of the payoffs.

Definition 26 (Monte Carlo price estimator) With payoff Φ and discount factor e^{-rT} ,

$$\hat{C}_N = e^{-rT} \frac{1}{N} \sum_{i=1}^N \Phi(S_T^{(i)}).$$

By the Law of Large Numbers $\hat{C}_N \rightarrow C$, and by the Central Limit Theorem the error is approximately normal, giving the confidence interval you built in MAT31-2:

$$\hat{C}_N \pm 1.96 \frac{\hat{s}}{\sqrt{N}},$$

where $\hat{s}$ is the sample standard deviation of the discounted payoffs. The $1/\sqrt{N}$ convergence is the defining feature — and the defining weakness — of Monte Carlo: to add one decimal digit of accuracy you need 100 times as many paths.

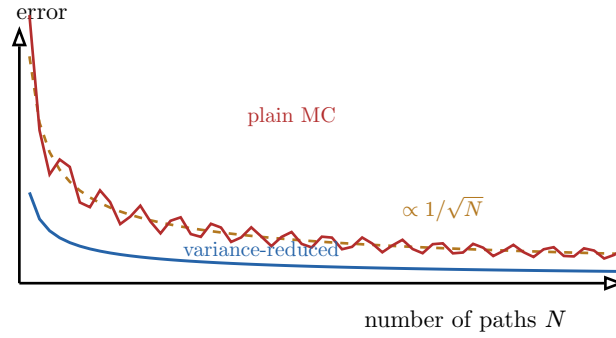


Figure 7: Monte Carlo error versus the number of paths N . Both the plain estimator (red) and a variance-reduced one (blue) decay like $1/\sqrt{N}$ (dashed envelope), but variance reduction lowers the **constant** — it shifts the whole curve down, buying the same accuracy with far fewer paths rather than changing the slow $1/\sqrt{N}$ rate.

7.3 Variance reduction

Because the error is $\hat{s}/\sqrt{N}$, halving the standard deviation $\hat{s}$ is worth quadrupling N — and it is usually far cheaper. Two classic techniques do exactly this.

Definition 27 (Antithetic variates) For each simulated path driven by normals $Z_1, \dots, Z_m$, simulate a **mirror** path driven by $-Z_1, \dots, -Z_m$, and average the two payoffs. Since Z and $-Z$ have the same law, the estimate stays unbiased; since the two paths are **negatively correlated**, the variance of their average drops below that of two independent paths.

Definition 28 (Control variates) Suppose a related quantity Y is simulated alongside the payoff X and its true mean $\mathbb{E}[Y]$ is **known** in closed form. Use the corrected estimator

$$X - c(Y - \mathbb{E}[Y]),$$

which has the same mean as X (because $\mathbb{E}[Y - \mathbb{E}[Y]] = 0$) but, for a well-chosen c , smaller variance — the known Y “explains away” part of the noise in X .

A natural control variate for an exotic option is the **underlying stock** itself, whose risk-neutral mean $\mathbb{E}^{\mathbb{Q}}[S_T] = S_0 e^{rT}$ is known exactly, or a vanilla call whose Black–Scholes price we have in closed form. Both techniques are unbiased: they reduce noise without shifting the answer, which is precisely what makes them safe to use.

Remark Variance reduction does not beat the $1/\sqrt{N}$ rate — that rate is a law of nature for plain Monte Carlo. What it buys is a smaller **constant**, often a factor of several, which on a large pricing run is the difference between minutes and hours. Quasi-Monte Carlo methods can improve the rate itself, but that lies beyond this course.

8 Epilogue: one rule, told seven ways

Look back over the journey. We began with a random walk whose displacement grew like $\sqrt{n}$, and that square root never left us. It became Brownian motion, whose increment over dt has size $\sqrt{dt}$ and whose square obeys $(dW)^2 = dt$. That one rule generated the Itô isometry (square a dW integral, get a ds integral), Itô’s formula (the $\frac{1}{2}f''\sigma^2 dt$ correction is the surviving second-order

term), the $-\frac{1}{2}\sigma^2$ in the solution of every log-driven SDE, the second-derivative term $\frac{1}{2}\sigma^2 S^2 V_{SS}$ in the Black–Scholes PDE, and the $\sqrt{\Delta t}$ scaling of every Euler–Maruyama simulation.

The other constant companion was the **martingale**, the fair game of MAT21-4 made formal. The Itô integral is a martingale because its integrand is fixed at the left endpoint; the zero-expectation property is that martingale property in disguise; risk-neutral pricing works because, under $\mathbb{Q}$, the discounted stock price is a martingale, so the option’s value today is simply a discounted expectation. “No improving direction left” was the refrain of the optimization course; “no expected gain from a fair bet” is the refrain of this one.

Carry both threads forward. Whenever you meet continuous-time randomness — in finance, in physics, in the diffusion models that now generate images and text — you will reach for the same two tools: **square the increment**, and **find the martingale**. Everything else is detail.